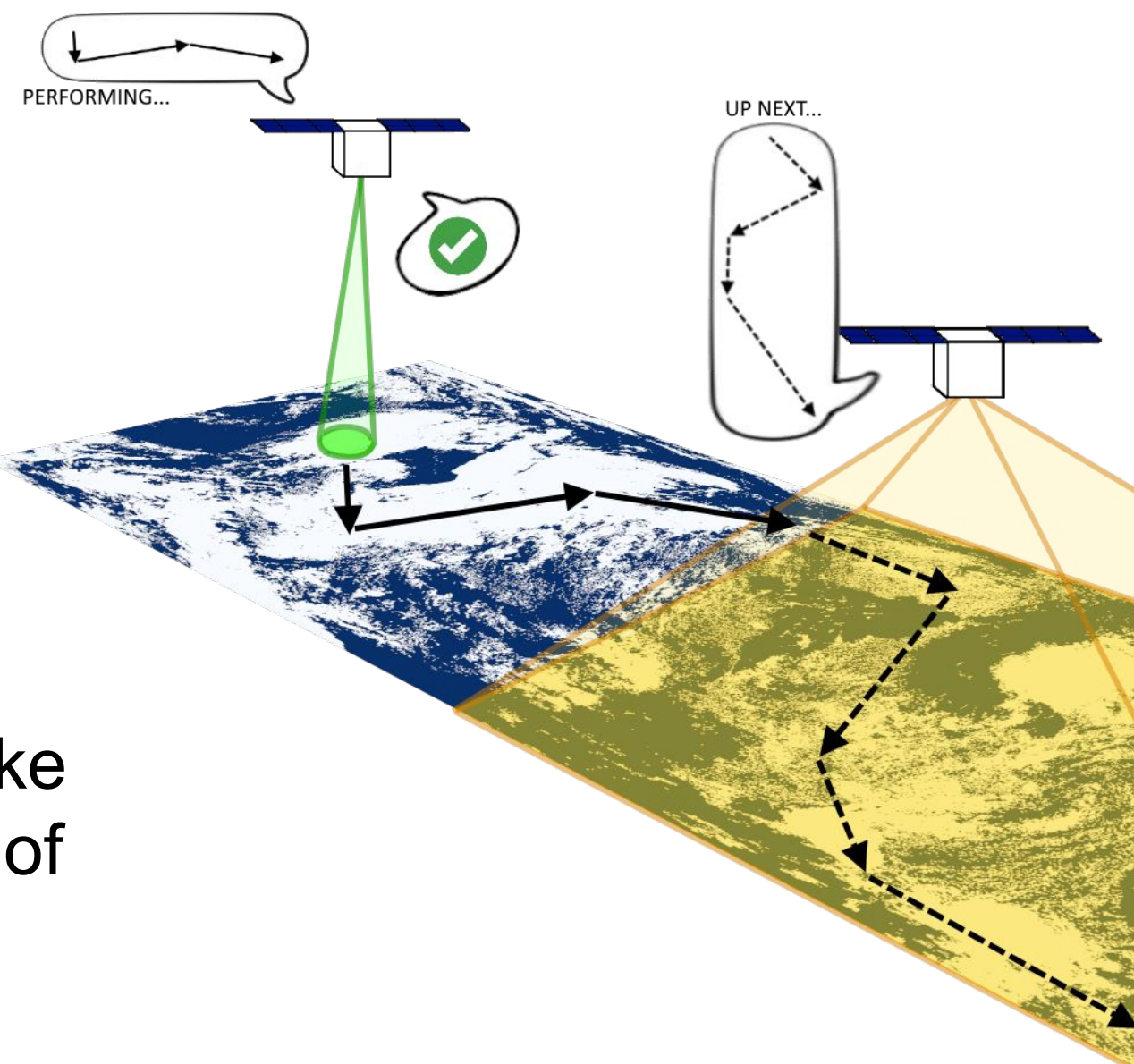




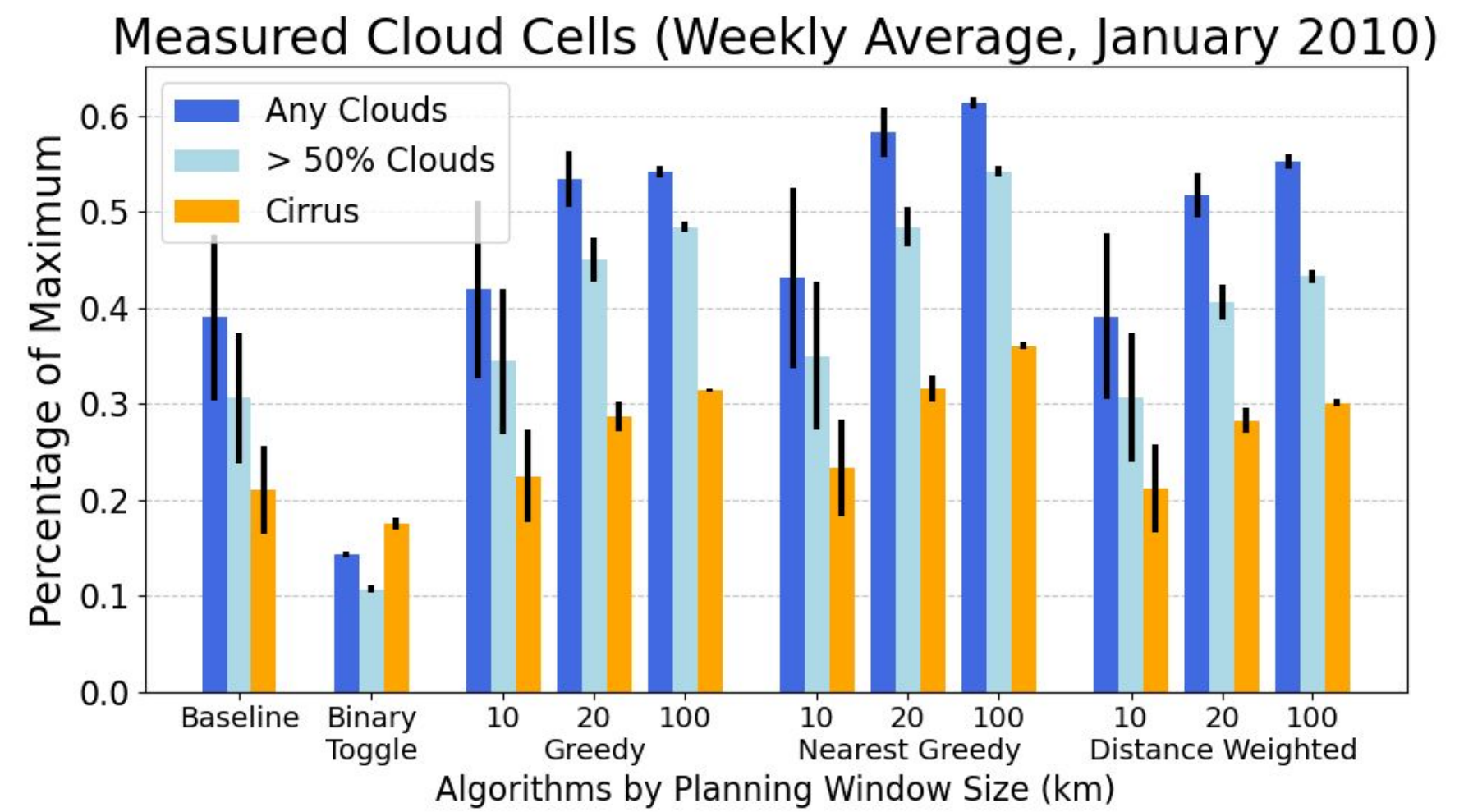
Introduction

Traditional cloud satellites waste **time** and **power** measuring empty sky, missing crucial, poorly understood clouds.

By using a **lookahead sensor**, a satellite can instead autonomously make **targeted measurements** of high value clouds [1].



Results

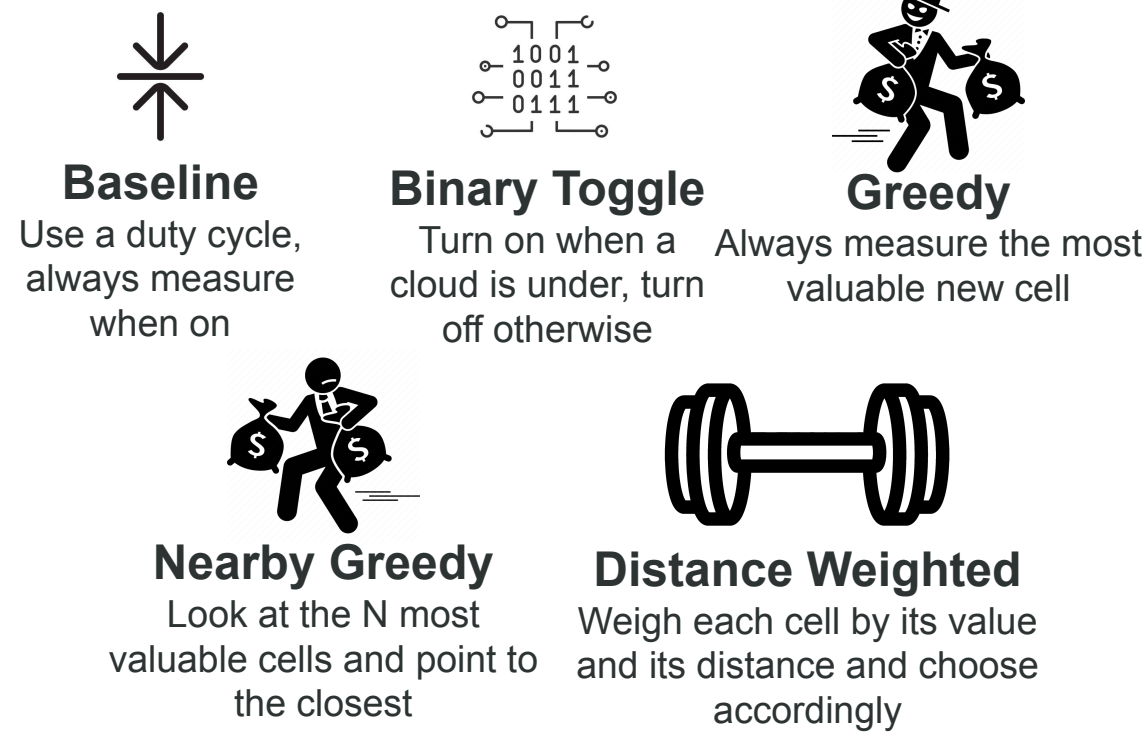
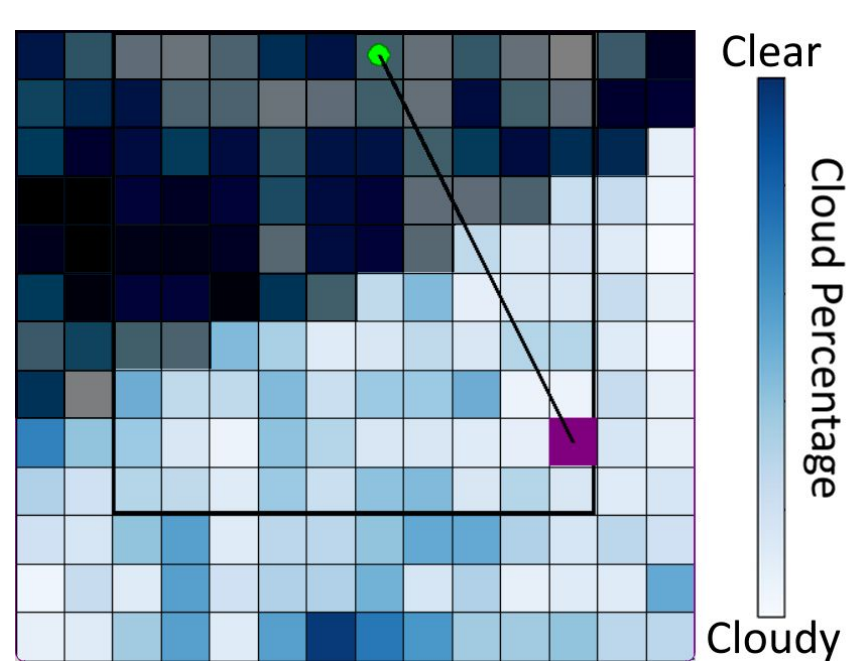


Algorithms with **20 km** or larger lookahead windows showed improvement over the baseline, both on average and for having less variance. **Nearest greedy** performed best.

Methods

Simulation for Algorithm Design

Using MODIS cloud masks, we built a simulation to test different prioritization algorithms for pointing at clouds.



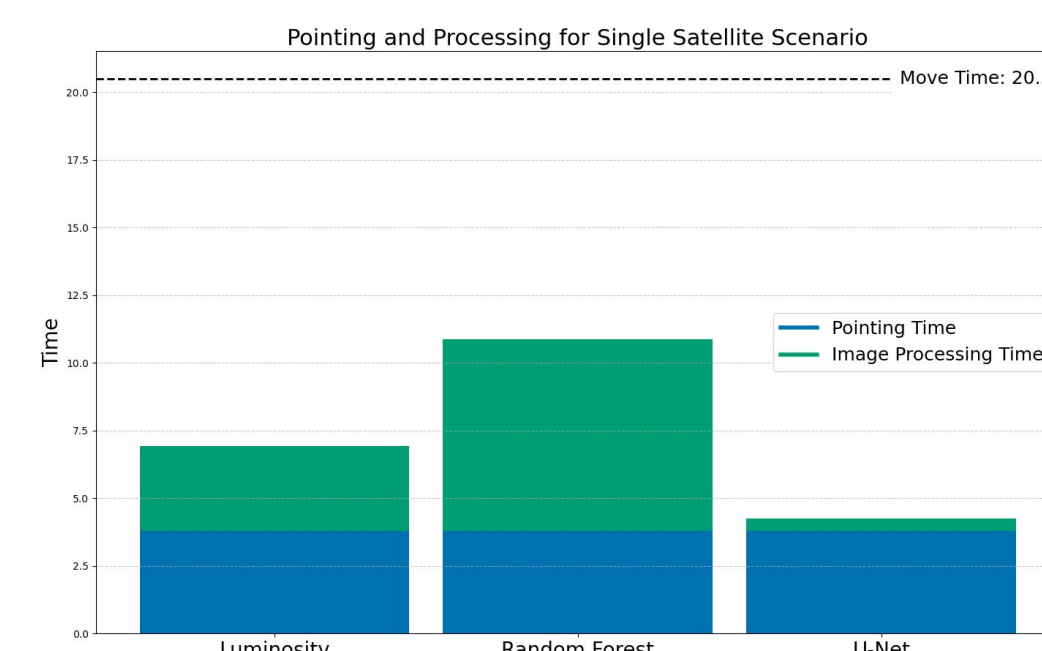
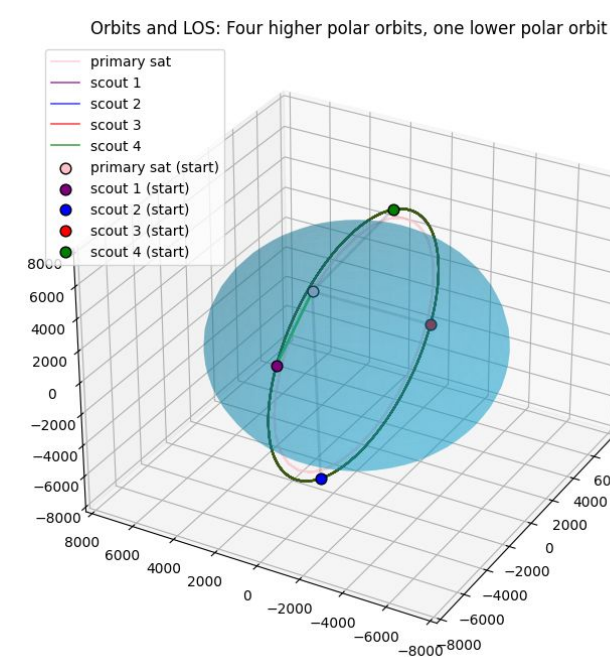
Lookahead Design

We analyzed strategies for taking lookaheads: a single satellite that points forward to image and scout satellites sending images to a main one.

Satellite Design

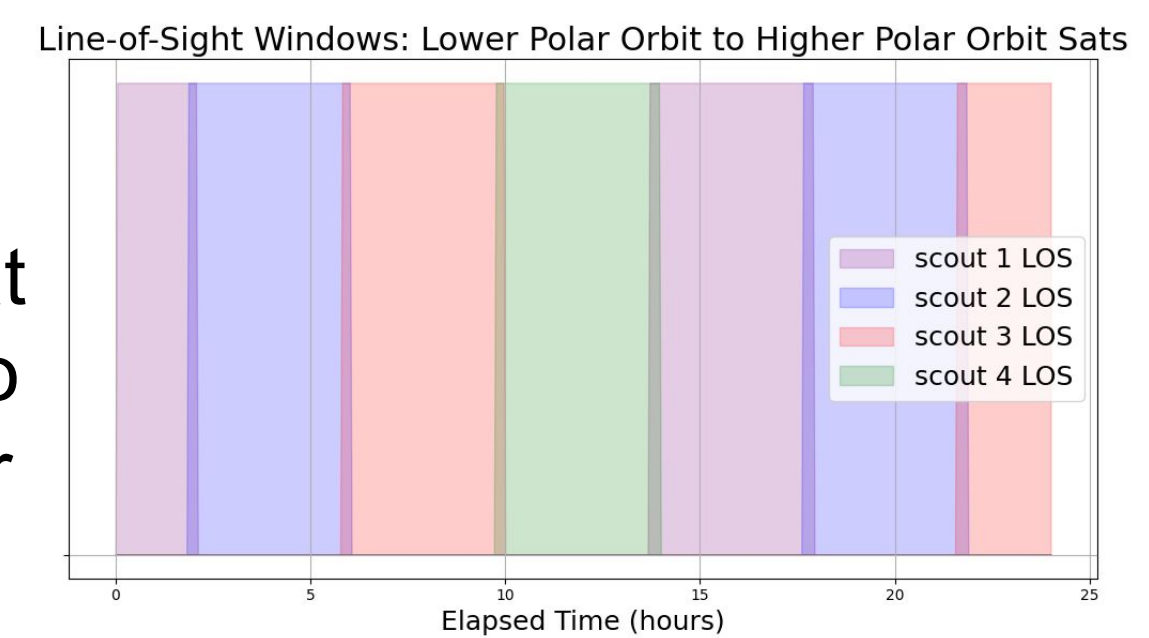
We identified high-value instruments for cloud science. We used a genetic algorithm to optimize “science per dollar.”

	CA	CTH	CTT	COD	AOD	LWP	r _e
Imager	1	0.6	0	0	0	0	0
Radar	0	0.6	0	0	0	0.4	0.4
Lidar	0	1	0	0.8	0.6	0.4	0.2
Radiometer	1	0.6	0.8	0.6	0	0.6	0.5
Polarimeter	0.2	0.8	0	0.4	0.8	0.4	0.4
MWR	0	0	0	0	0	0.8	0.2

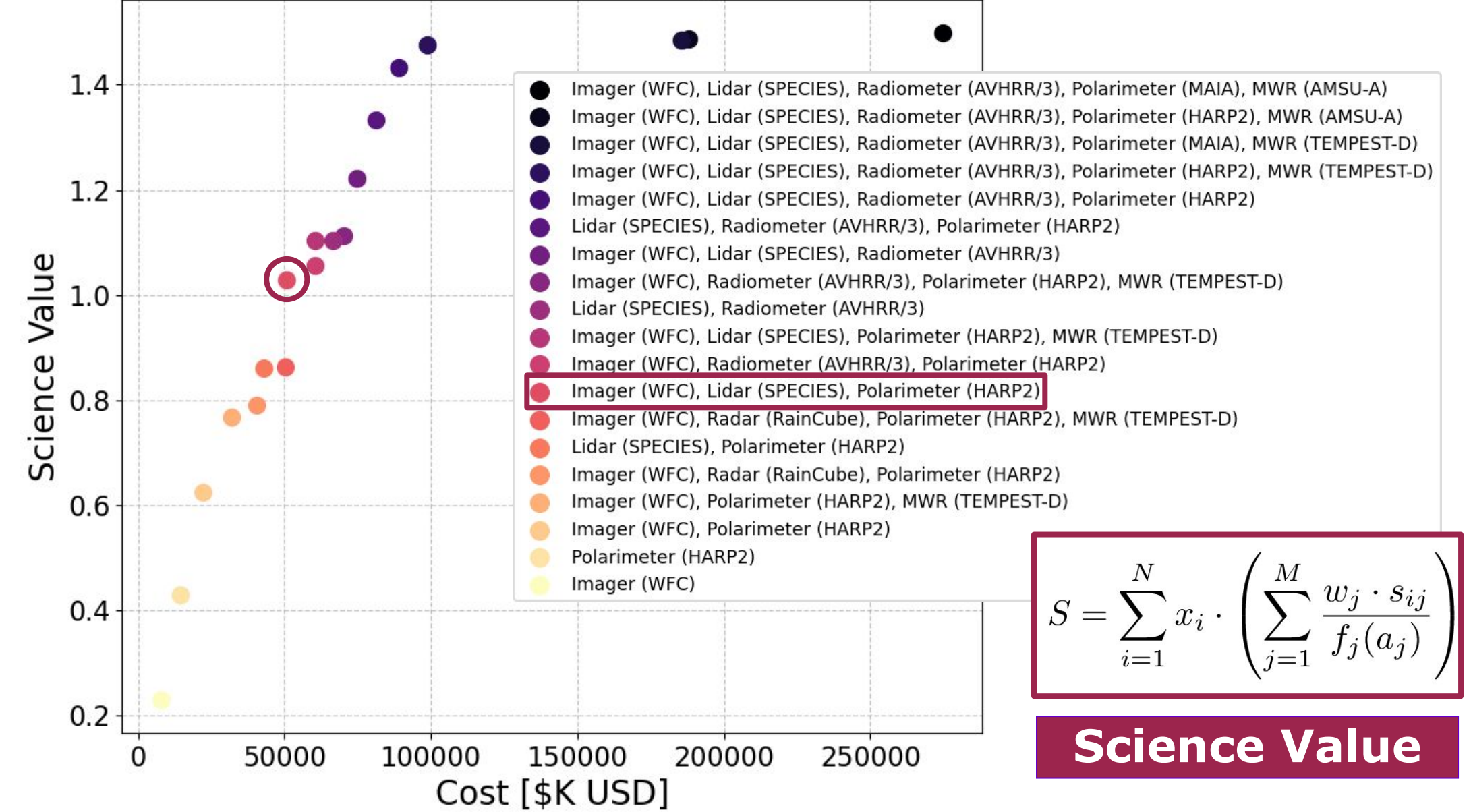


For a **single satellite** design, there is sufficient time to point forward and process a cloud image [2] before the satellite passes that area.

There are many configurations of scout satellites in **higher orbits** that offer consistent line of sight to the main satellite, allowing for larger **lookahead images**.



NSGA2 Pareto Front, Discovery Class Mission



Polarimeters are frequently included in optimal designs.

Conclusions

Algorithm Performance

Prioritization algorithms need to balance **measuring high value cells** and **limiting pointing**.

Lookahead Configurations

Viable designs are **single satellite**, **one scout in the same orbit**, **four high-LEO scouts**, and **two GEO scouts**.

Satellite Instrumentation

The **imager, lidar, polarimeter** combination design best balances science value per dollar and will function in the dynamic framework.

Future Work

Simulation: Simulate the requirements for specific **instruments**, increase the number of **cloud targets**, implement **time to receive lookahead** to verify lookahead and satellite instrument choices.

Lookahead: Model **communications** and **image distortion**.

Satellite system design: Assess capabilities of instruments to measure **off-nadir** and penalize certain **combinations**, depending on capabilities.

References

- [1] Dahl, Cahoy, “Improving cloud observations by autonomously pointing satellites,” *to appear in IEEE IGARSS*, Aug. 2025
- [2] Kacker, Meredith, Kusters, Tornio, Felt, Cahoy, “On-orbit rule-based and deep learning image segmentation strategies,” *AIAA SCITECH*, Jan. 2022.